Unit -3 Advanced Engineering Materials.

polymers:- polymers are the high molecular weight compounds obtained by repeated union of simple molecules. Ex:- Starch, polyvinyl chloride, polyethylene.

➤ **Monomers:-** Monomer is a simple repetitive unit which when joined together in large numbers which give rise to a polymer. These are the building blocks of polymer.
Ex:- Vinyl chloride, ethene, propylene etc.

- The number of repeating units (n) in the chain so formed is called the Degree of polymerization polymers with a high degree of polymerization are called High polymers and those with low degree of polymerization are called Oligopolymers.
- Degree of polymerization is a number. which indicates the number of repetitive units present in the polymer.

Ex:- $(CH_2-CH_2)n$ polythene

① Calculate the molecular weight of the polythene polymer given DP is 100.
Molecular weight of the polythene
= DP x Molecular weight of polethene
= 100 x 28 = 2800.

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1] Functionality:-
- The number of bonding sites (or) reactive sites or functional groups present in the molecule.

2] Nomenclature of polymers:-
- There are four types.

a] Homo polymers:- polymers made up of with ~~different~~ type of monomers are called ~~hetero~~ homo polymers. e.g:- polyethylene, PVC.

b] Hetero polymers:- polymers made up of with different type of monomers are called hetero polymers. e.g:- buna-s-rubber.

- Graft copolymers:- If main chain consists of one monomer and branched chain consists of another monomers are called Graft copolymers.

c] Homo chain polymers:- If the main chain consists of only one type of atoms called homo chain polymers. e.g:- polyethylene.

d] Hetero chain polymers:- If the main chain consists of different type of atoms called hetero chain polymers. e.g:- polyester, Nylon-6.

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- Copolymers:- Homopolymers from a single monomer. copolymer from two or more monomer

3) Tacticity:- The difference in configuration due to the orientations of different functional groups with respect to the main chain is called tacticity. It is 3 types.

a) Isotactic:- If all the functional groups are arranged on the same side of main chains is called isotactic.

b) Syndiotactic:- If all the functional groups are arranged in the alternative fishion of main chain is called syndiotactic.

c) Atactic:- If all the functional groups are arranged in random of main chain is called atactic.

Basis of classification	polymer Type
- Origin	- Natural, semisynthesis, synthesis.
- Thermal Response.	- Thermoplastic, thermo-setting.
- Mode of formation	- Addition, Condensation.
- Line structure	- Linear, Branched, Cross-linked.
- Application & physical properties	- Rubber, plastic, fibers

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- Tacticity - Isotactic, syndiotactic, Atactic
- Crystalinity - Non crystalline (amorphous), semi-crystalline, crystalline

4) Classification Based on Source.

a) Natural polymer :- These polymers are found in plants and animals. Examples:- proteins, cellulose, starch.

b) Semi-synthetic polymer :- Cellulose derivatives as cellulose acetate and cellulose nitrate. etc.

c) Synthetic polymer:- A variety of synthetic polymers as plastic, synthetic fibers and synthetic rubbers.

5) classification Based on Backbone of the polymer chain.

a) Organic and polymer:- A polymer whose backbone chain is essentially made of carbon atoms is termed as organic polymer.

b) Inorganic polymer:- The chains backbone contains no carbon atom is called Inorganic polymer.

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5) Classification Based on structure of polymers.

a) Linear polymer:- The polymer consist of long and straight chain. ex:- HDP, PVC etc

b) Branched polymer:- The polymer contain linear chain having some branches. ex:- LDP.

c) Cross-linked polymer:- There are usally formed from bi-functional and tri-functional monomers and contain strong covalent bonds between various linear polymer chains. ex:- vulcanized rubber, urea formldehyde resine.

6) Classification Based on Composition of polymer:-

a) Homopolymer:- A polymer resulting from the polymerization of a single monomer, a polymer consisting substantially of a single type of of repeating unit.

b) Copolymer:- When two different types of monomers are joined in the same polymer chain, the polymer is called copolymers

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7) Classification Based on Mole of polymerisation:-

a) Addition polymers:- The addition polymers are formed by the repeated addition of monomers molecules possesing double or triple bond.

e.g.:- polythene.

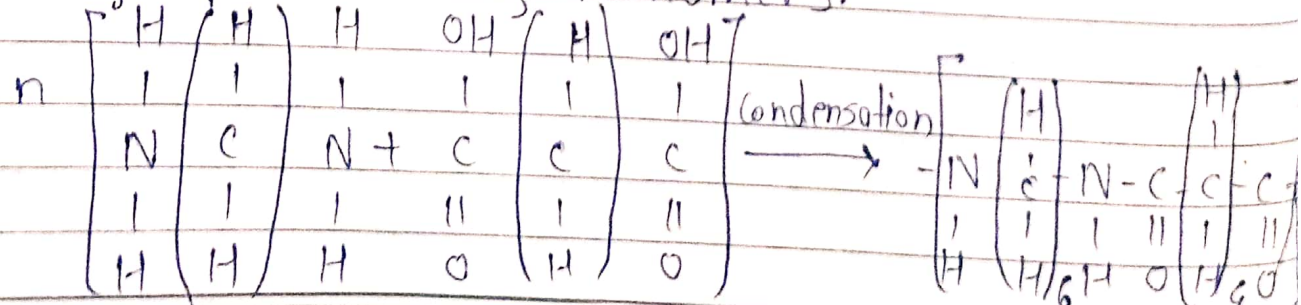
$$n CH_2 = CH_2 \longrightarrow -(CH_2 - CH_2)_n$$ Homopolymer
polythene

- The polymers made by addition polymerisation from two different monomers are termed as copolymers. eg:- Buna-S, Buna-N etc.

$$n(CH_2 = CH - CH = CH_2 + n(C_6H_5CH = CH_2$$
1, 3-Butadiene styrene

$$\longrightarrow -((CH_2 - CH = CH - CH_2 - CH_2 - \underset{C_6H_5}{CH})_n$$
Butadiene-styrene copolymer (Buna-S).

b) Condensation polymers:- It is defined as a reaction occuring between simple polar-group-containing monomers.



Addition polymerization	Condensation polymerization
1) Only growth reaction adds repeating units one at a time to the chain.	1) Any two molecular species present can react.
2) Number of units decreases steadily throughout the reaction.	2) Monomer disappears early in the reactions.
3) High molecular mass polymer is formed at once.	3) polymer molecular mass rises steadily throughout the reaction
4) Longer reaction times have a little effect on molecular-weight but gives higher yields.	4) To obtain high molecular-weight, longer reaction time is essential.
5) The reaction mixture contains only monomers, high polymers & about 10^{-8} part of growing chains.	5) All types of molecular species are present at any stage.

8) Classification based on molecular forces:-

a) Elastomer:- The polymer chains are held together by the weakest intermolecular forces. ex:- buna-S, buna-N, neoprene.

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b) Fibers :- Fibers are the thread forming solids which posses high tensile strength and high modulus. ex:- polyamids, polyesters.

c) Thermoplastic polymers :- Some polymers soften on heating and can be converted into any shape that they can retain on cooling.

d) Thermosetting polymer :- Some polymers on the other hand, undergo some chemical change on heating and convert themselves into an infusible mass.

- Solids show a wide range of electrical conduction from 10^{-20} to 10^{7} $ohm^{-1} m^{-1}$. On the basis of electrical conductivity, the solid can be broadly classified into three types.

1) Conductors :- The solids which have conduction in order of 10^{4} to 10^{7} $ohm^{-1} m^{-1}$. Metals are good conductors of electricity.

2) Insulators :- Solids which have very low conductivity in the range 10^{-20} to 10^{-10} $ohm^{-1} m^{-1}$.

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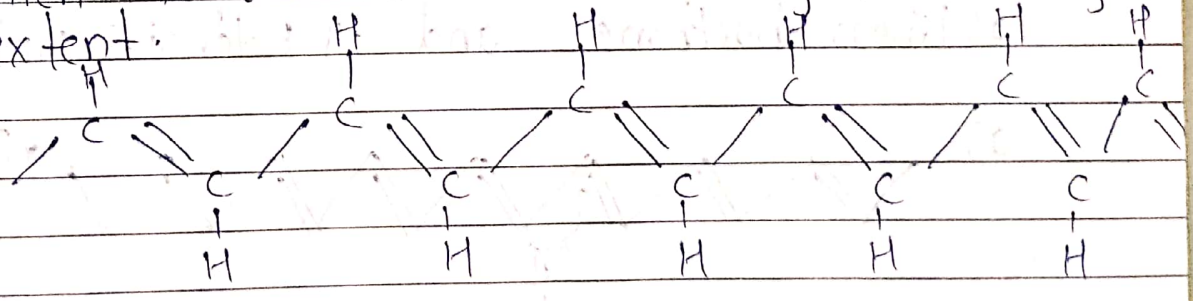
3) Semiconductors :- Their conductivity is in between conductors and insulator up to the order of 10^{-6} to 10^{4} $ohm^{-1} m^{-1}$.

Band Theory :- The difference between valence band and conduction band is called band gap.

- Valence band :- These is a band of lower energy.
- Conduction band :- These is a band of higher energy.

Intrinsically Conducting polymers :-

- Extensive conjugation in the backbone which is responsible for conductance.
- Contain conjugated pie-electron in the backbone which increase their conductivity to a large extent.



Conducting polymers :-

- The polymers which conduct the electricity are called conducting polymes.

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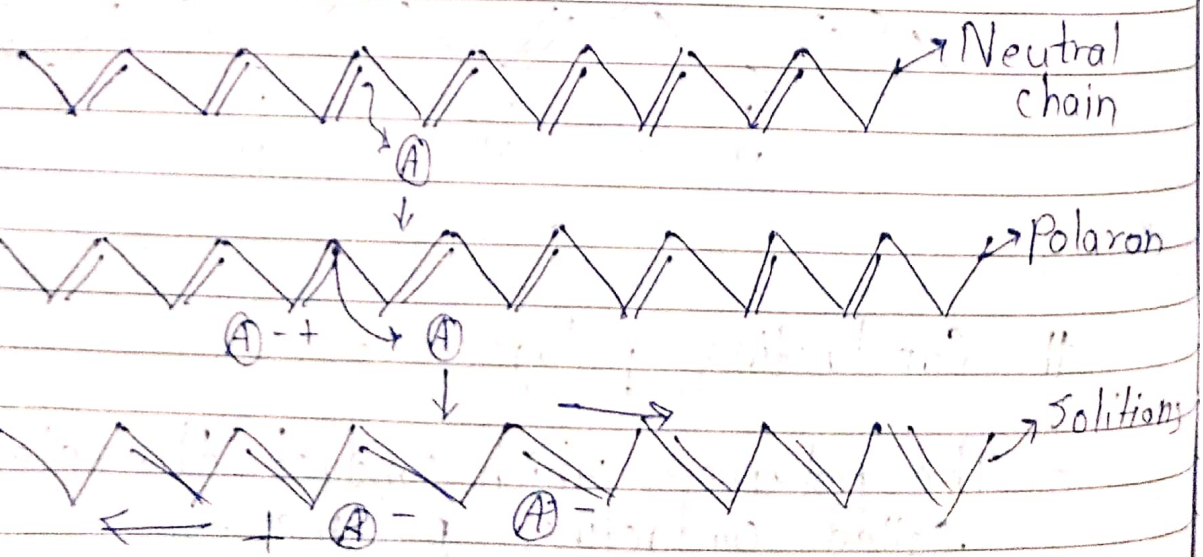
Conducting polymers

- Intrinsically Conducting polymers
  - Conducting polymers having conjugated π-electrons in backbone
  - Doped conducting polymers
    - p-Doping
    - n-Doping
- Extrinsically Conducting polymers
  - Conductive element Filled polymers
  - Blended conducting polymers
- Co-ordination conducting polymers

Conducting Mechanism:-

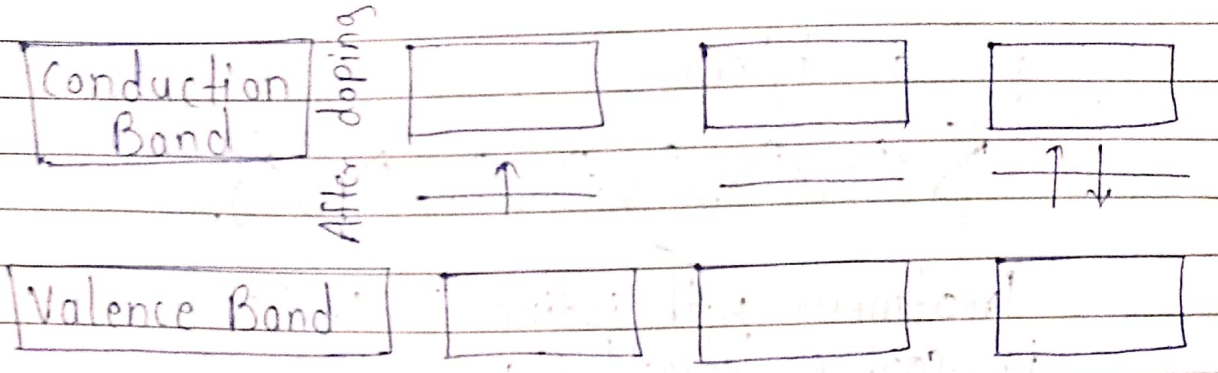
Requirements of conducting polymer:
1) linear backbone and 2) Extensive conjugation.

a)



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b)



pristine trans-polyacetylene | Neutral Solition | Positively charged Solition | Negatively charged Solition.

- ICP posses conjugated pi-electrons in backbone when such polymer faces electrical field pi-election get existed and mores through it polymaric material.
- The orbitals of conjugated pi-electrons get overlap on the backbone and hence valence band and conduction bands are develop which get distributed over intire surface of polymer appropriate conduction of pi-electron makes polymer to conduct electricity very efficiency.

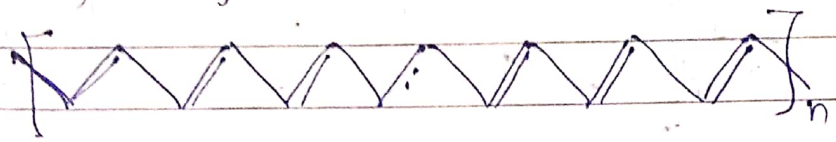
The requirements for a conducting polymer for electrical conduction are.

1) The molecule should have a linear backbone.
2) The molecules should have extended conjugation.
3) charge carries (either positive holes or electrons) should be introduced by appropriate dopants.

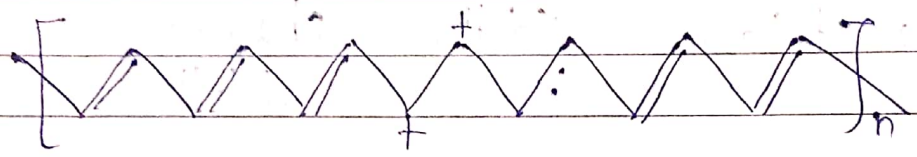
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polyacetylene

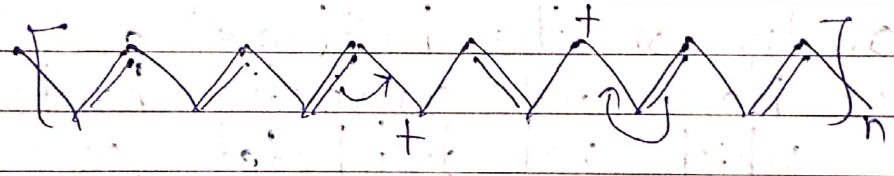


Treatment with iodine I_2/CCl_4
in carbon tetrachloride

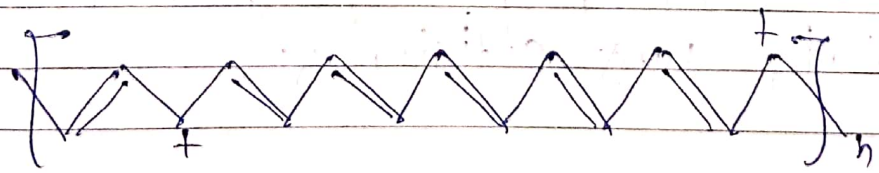


polaron

Treatment with iodine I_2/CCl_4
in carbon tetrachloride



Bipolaron



Soliton

CB LUMO polyacetylene is a neutral molecule with extended conjugation. It's conductivity is very low. On doping with suitable dopants, conductivity increases.

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VB] HOMO

CB] LUMO

—

+↑

polaron can be a radical cation or radical anion. In this case polaron is a radicle cation.

VB] HOMO

CB] LUMO

++

Bipolaron of polyacetylene in this mechanism contains two positive charges (dication)

VB] HOMO

CB] LUMO

+ +

Further delocalization of charges can be observed

- polyacetylene in the first example of synthetic conducting polymer.
- Acetylene on passing over Zeiglar Natta catalyst at room tempreature forms polyacetylen.
- $CH \equiv CH \xrightarrow{\text{Zeiglar Natta}} CH = CH - CH = CH$
  Acetylene → polyacetylene
- polyacetylene film containing a mixture of both cis and trans structures is formed cis from is copper in colour while trans structure silvery.

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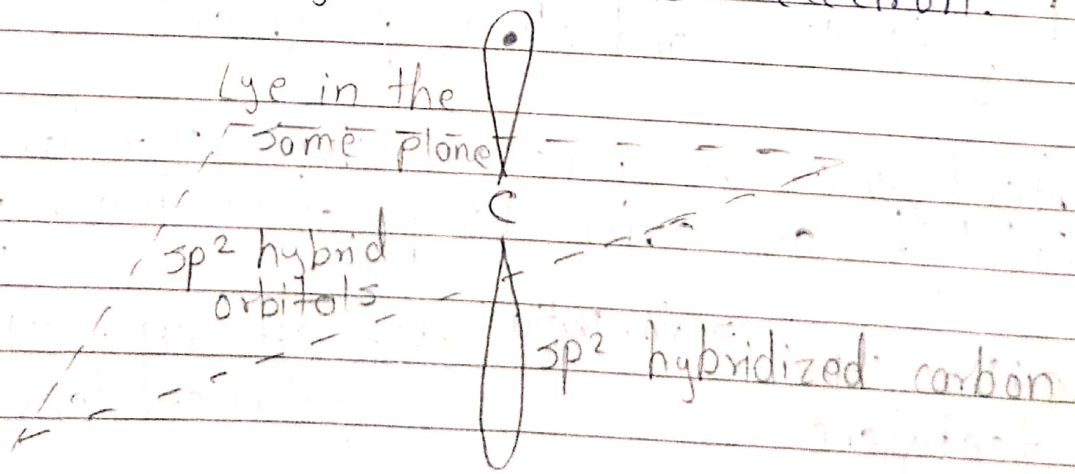
P - doping

$[CH]_x + 3x[2I_2] \longrightarrow [CH]_x^{x+} + xI_3^-$

N - doping

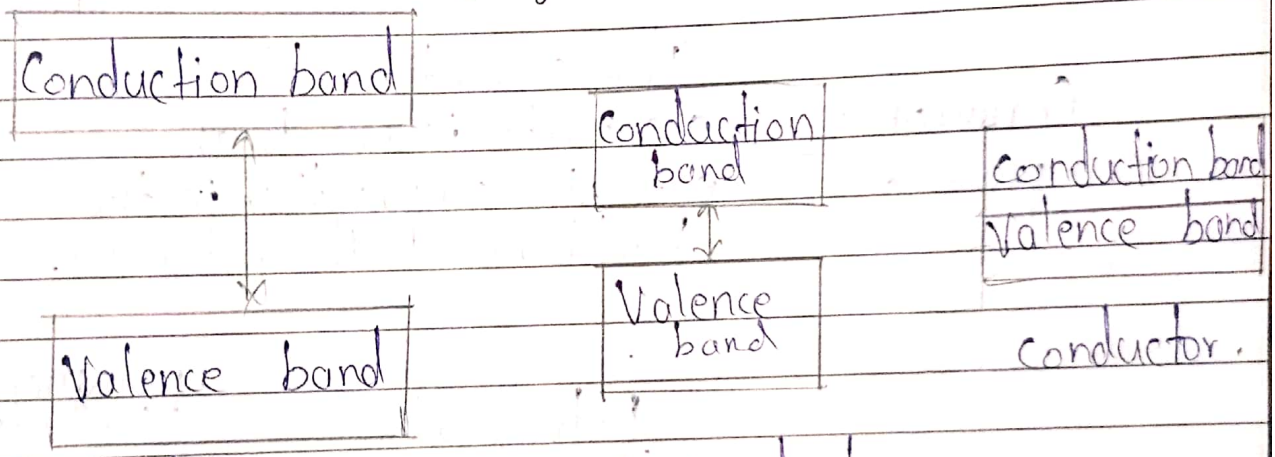
$[CH]_x + xNa \longrightarrow [CH]_n^{x-} + xNa^+$

- In a conducting polymer, sp^2 hybridized carbon chain constitute the backbone for conduction.
- On each sp^2 hybridized carbon, there will be a single electron in the unhybridized p_z orbital. The unhybridized p_z lies orthogonal to sp^2 hybridized orbitals.
- The delocalization of electrons present in the p_z orbitals results in conduction.
- Unhybridized p orbital lying orthogonal to sp^2 hybrid orbitals electron.



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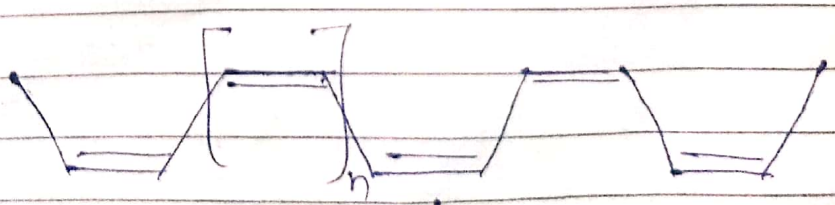
- According to band theory, in the case of insulator, between the valence band and the conduction band the forbidden gap is more, while in the case of semiconductor the forbidden gap is moderate. In the case of conductors, the forbidden gap is negligible.
- In conjugated polymers, the forbidden gap can be decreased by introducing charge carries through doping. This process facilities conduction of conjugated polymers.



Cis polymers:-

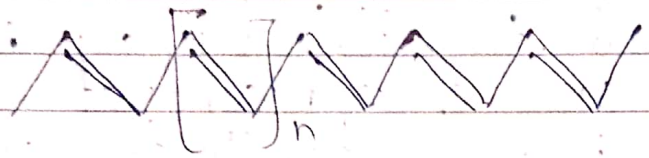
At lower temperature (-80°C) cis form exists.

Cis form on heating above 150°C, converts to trans structure.



cis polyacetylene

Trans polymers:-



Trans - polyacetylene.

Conducting polymers

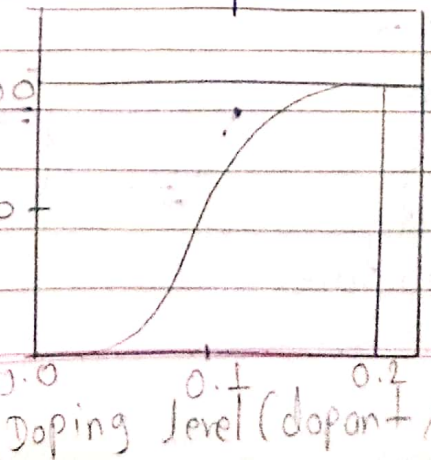
- Intrinsically conducting polymers
  - Conjugated conducting polymers
  - Doped conducting polymers

The conduction of electricity is due to the extensive conjugation in the backbone of the polymers.

- Extrinsically conducting polymers
  - Polymers with conducting elements
  - Polymers with conducting blends.

plot of conductivity vs doping

Conductivity increases up to a certain doping level.



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Advantages :-

- Used because of either their light weight, biological compatibility..
- Increase speed and reduce power consumption.
- We can make flexible transparent displays.

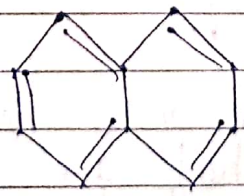
Conductivity polymers :-

- Conduction mechanism in polyacetylene.

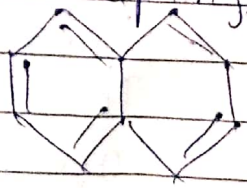
① Unsatured polymers consists of hydrogen and sp_2 hybridized carbon atoms.
② The localized electrons from the backbone of polymer chain and dominate the mechanical properties.
③ The electrons in the pi bond are delocalized along the chain and are responsible for optical and electrical properties of conjugated polymers.

2-Reduction with lewis Base (called-ndoping)

polymer + lewis Base ⟶ n-Doped polymer (reductive coupling)



$(CH)_x$ + Na^+ ⟶ $Na^+(CH)^-_x$ +

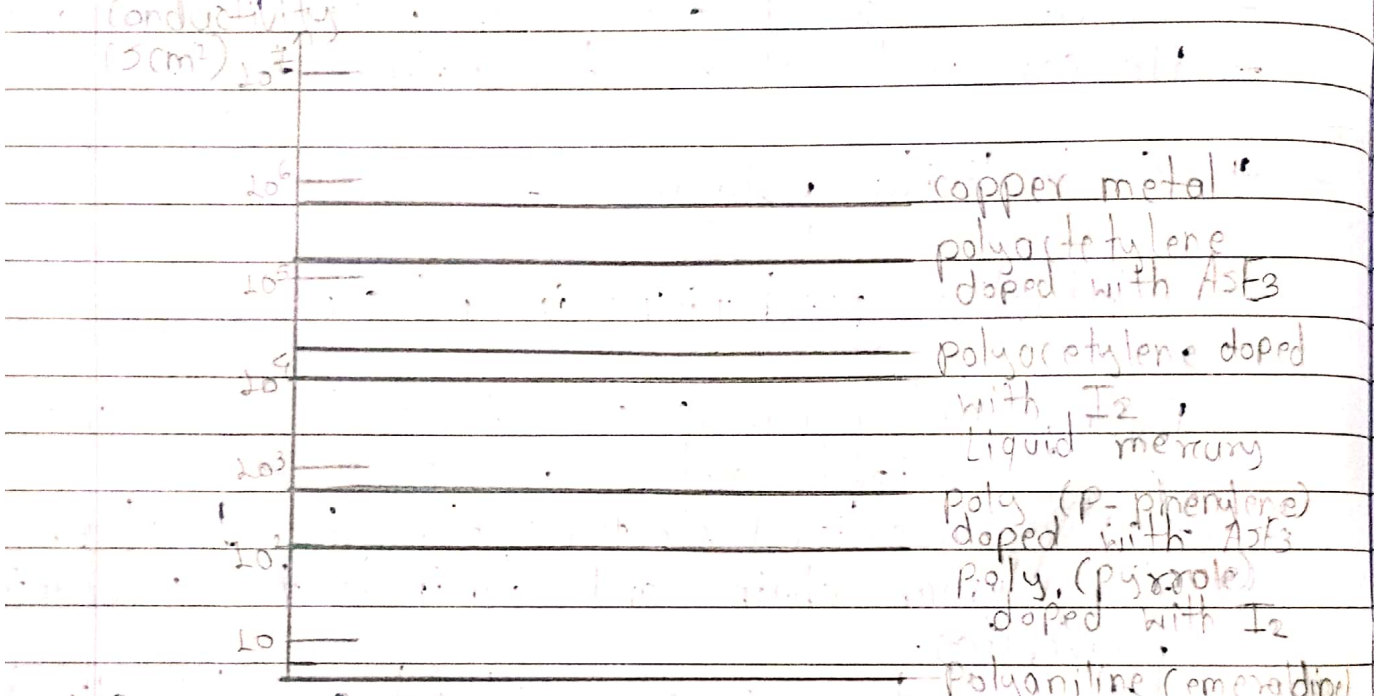


polyacetylene Sodium napthilide n-doped polyacetylene napthalene.

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Conducting polymers:-

- Comparison of conductivity with metals.



Biodegradable polymers:-

- Biodegradable polymers are defined as polymers combined of monomers linked to one another through functional groups and have unstable links in the backbone
- They are broken-down into biologically acceptable molecules that are metabolized and removed from the body via normal metabolic pathways.
- Based on biodegradability polymers are classified are:-

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1) Biodegradable polymers.

- e.g:- collagen, poly glycolic acid etc.

2) Non biodegradable polymers.

- e.g:- poly vinyl chloride, polyethylene etc.

Polymer Degradation:-

- Polymer degradation is a change in the properties - tensile strength, colour, shape, etc. of a polymer or polymer based product under the influence of one or more environmental factors such as heat, light or chemicals.
- The term 'biodegradation' is limited to the description of chemical process (chemical changes that after either the molecular weight or solubility of the polymer).
- Bit 'Bioerosion' may be restricted to refer to physical process that results in weight loss of a polymer device with respect to time.

Natural polymers:-

- The use of natural biodegradable polymers to deliver drugs continues to be an area of active research despite the advent of synthetic biodegrable polymers.

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- Natural polymers remain attractive primarily because,
- They are an attractive class of biodegrable polymers.
- They are derived from natural sources.
- They are easily available.
- They are relatively cheap.
- They qualify for a number of chemical modifications.
- They can be a protein or a polysaccharide in chemical origin.

Biodegradable polymers can be classified in two on the basis of their sources.

➤ Natural polymers:-

- proteins:- Example = Albumin, Collagen, Gelatin etc.
- polysaccharides:- Example = Sodium alginate, chitin, chitosan, Cellulose, Dextran, Insulin, starch etc.

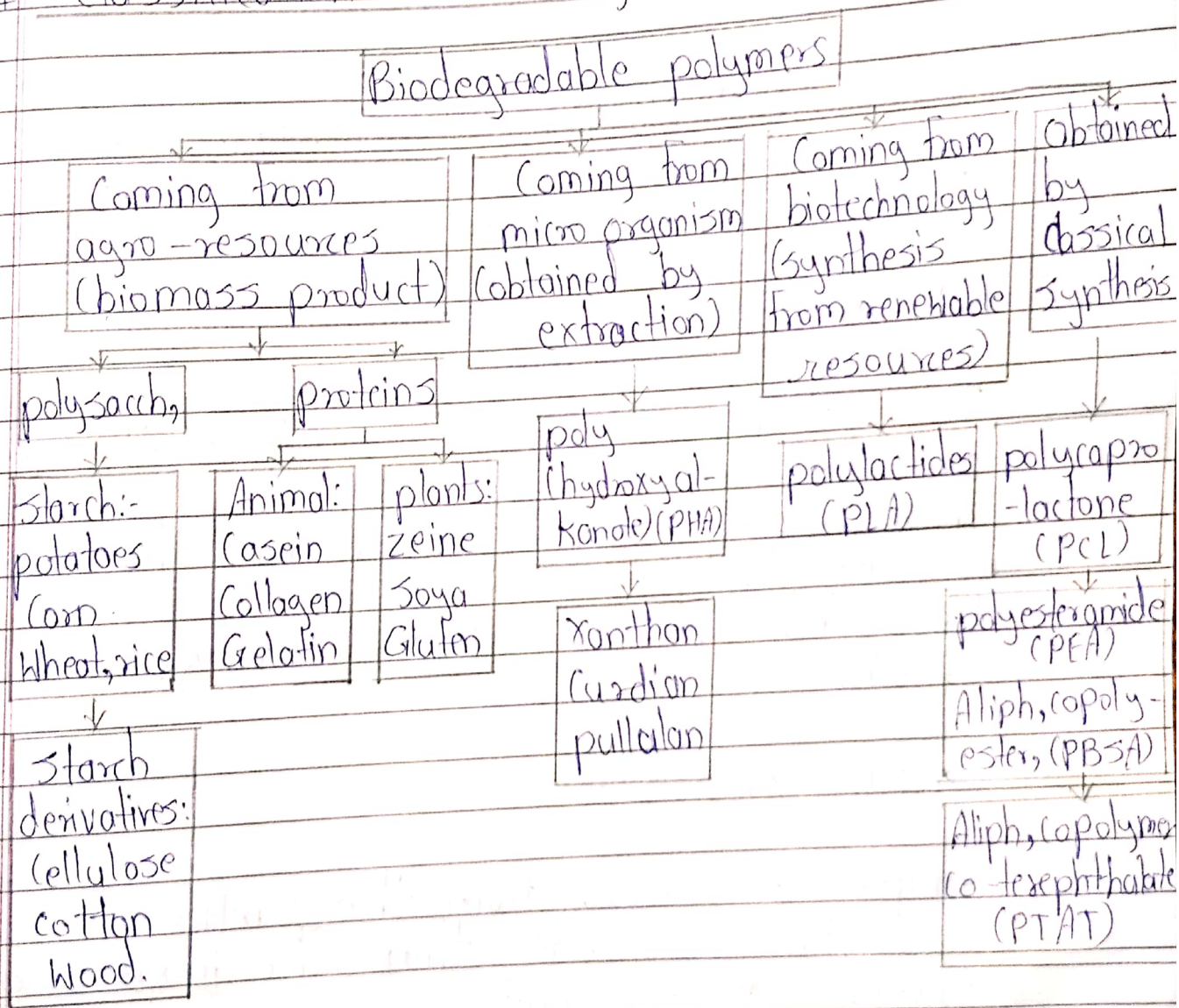
➤ Synthetic polymers:-

- Aliphatic polyesters:- Example = poly-Glycolic acid (PGA), poly Lactic Acid (PLA), poly-Hydroxy Butyrate (PHB), poly-β malic acid (PMA) etc.

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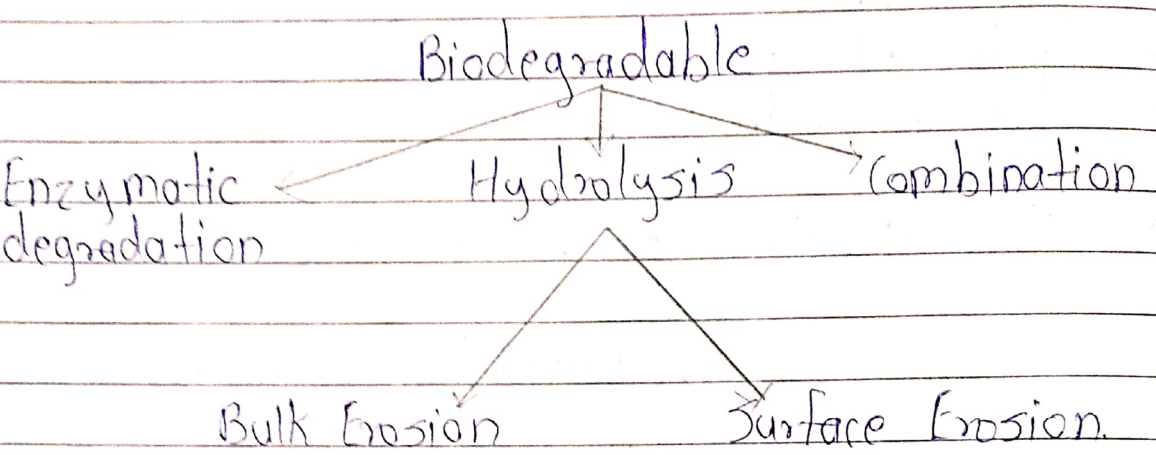
- poly phospho Esters.
- poly Anhydrides.
- poly phosphazenes.
- pseudo Amino Acids.

Classification of biodegradable polymers:-



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Mechanism of Biodegradable polymers:-



Types of bioerosion:-

① Bulk erosion:-

- Degradation takes place throughout the whole of the sample.
- Ingress of water is faster than the rate of degradation.
- E.g.:- polylactic acid (PLA), polyglycolic acid (PGA).

② Surface erosion:-

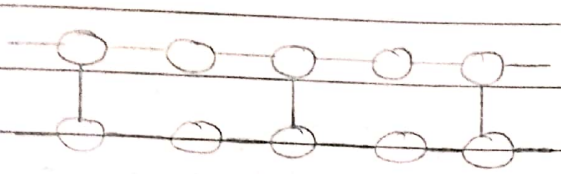
- Sample is eroded from the surface.
- Mass loss is faster than the ingress of water into the bulk.
- E.g.:- polyanhydrides.

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Enzymatic or chemical Degradation:-

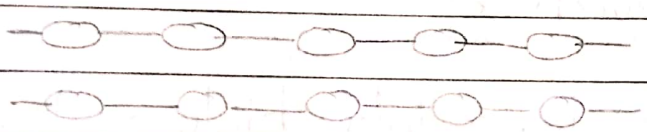
- chemical or enzymatic degradation - It is mediated by water, enzymes, microorganisms.

cleavage of CROSSLINKS.



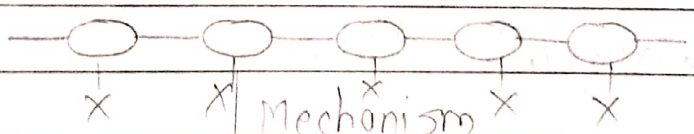
Water Insoluble

Mechanism
cleavage of crosslinks between water soluble polymer chains



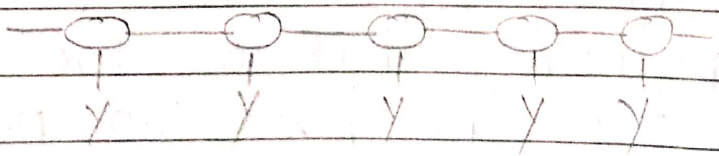
Water soluble

Transformation of side chains.



Water Insoluble

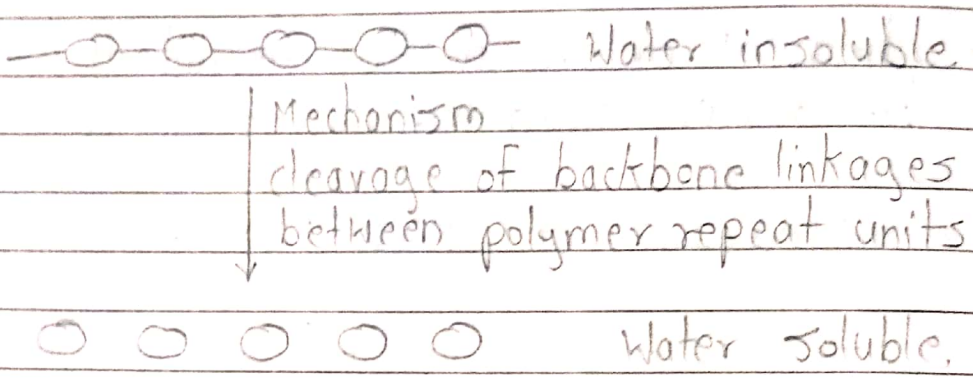
Mechanism
Transformation or cleavage of side chain leading to the formation of polar or charged group.



Water soluble.

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cleavage of Backbone.



- They slowly disappear from the site of administration in response to a chemical reaction such as hydrolysis.

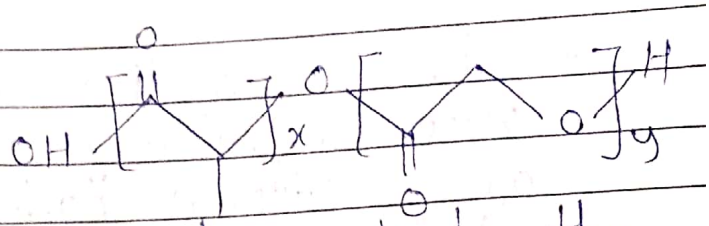
Biodegradable polymer:-

- They are broken down into biologically acceptable molecules that are metabolized and removed from the body via normal metabolic pathways.
- Example:- polylactic Acid.
  polyglycolic Acid.

- Biodegradable polymers are natural or synthetic in origin and are degraded in vivo, either enzymatically or non-enzymatically or both.
- They are broken down into biologically acceptable molecules that are metabolized and removed from the body via normal metabolic pathways.

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Example:-
polylactic Acid
polyglycolic acid
chitosan.

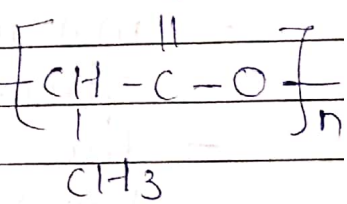


x and y indicate the no. of times each unit repeats.

poly (lactic-co-glycolic acid)

General properties of polylactic acid:-

- Transperancy
- Glass Transition tempreature 50-60°C
- Melting point 130-180°C
- Crystallinity: 35-40 %.
- Tensile strength: 4-6 kg/mm²
- Elongation: 3-4 %, brittle.
- Flexural strength: 9-11 kg/mm²



Based on type of degradation.

- Which contains both physical (dissolution) and chemical (backbone cleavage) process.
chemical degradation.

- Mechanism of Degradation.
  - Bioerosion
  - Surface erosion.
  - chemical Degradation.

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- Slow dissolution and erosion by hydrolysis water insoluble polymers degradation of pendent group without backbone cleavage.
- Water insoluble polymer degrades to water-soluble products by backbone cleavage.
- H_2O soluble Swelling Dimensional stability H_2O insoluble chemical change No backbone cleavage H 20 insoluble chemical cleavage MW

1] Collagen:-

- Collagen is the most widely found protein in mammals and is the major provider of strength to tissue.
- The number of biomedical applications in which collagen have been utilized is too high. it not only has been explored for use in various types of surgery, cosmetics and drug delivery, but also in bioprosthetic implants and tissue engineering of multiple organs as well.
- It is used as sutures, Dressings, etc.

➤ Disadvantages:-

- poor dimensional stability. Variability in drug release kinetics.
- poor mechanical strength.

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2) Albumin :-

- It is a major plasma protein components.
- It accounts for more than 55% of total protein in human plasma.
- It is used to design particulate drug delivery systems.

➤ Applications :-

- Albumin micro-spheres are used to deliver drugs like Insulin, Sulphadiazene, 5-Fluoro-uracil, prednisolone etc.
- It is mainly used in chemotherapy, to achieve high local drug concentration for relatively longer time.

3) Dextran :-

- Dextran is a complex branched polysaccharide made of many glucose molecules joined into chains of varying lengths.
- It consists of α-D-1, 6-glucose-linked glucan with side-chains linked to the backbone of polymer. Its Mol. wt ranges from 1000 to 2,00,000 Daltons.

➤ Applications :- used for colonic delivery of drug in the from of gels.

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4) Gelatin:-

- Gelatin is a mixture of peptides and proteins produced by partial hydrolysis of collagen, extracted from the boiled bones, connective tissues, organs and some intestines of animals.

Factors Affecting Biodegradation of polymers:-

➤ Morphological factors:-
- Shape and size.
- Variation of different coefficient and mechanical stresses.

➤ Chemical factors:-
- Chemical structure and composition.
- Presence of ionic group and configuration structure.
- Molecular weight and presence of low molecular weight compounds.

➤ physical factors:-
- processing condition.
- Sterilization process.

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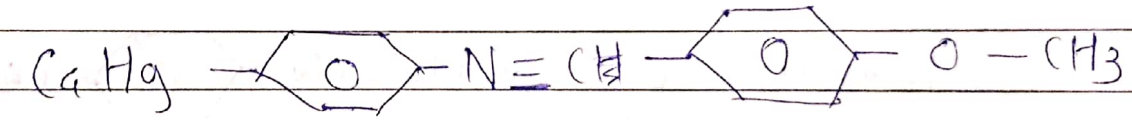
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Advantages of Biodegradable polymers:-

- Localized delivery of drug.
- Sustained delivery of drug.
- Stabilization of drug.
- Decrease in dosing frequency.
- Reduce side effects.
- Improved patient compliance.
- Controllable degradation rate.

Liquid Crystals [LCP] polymers [LCP] :-
(liquid crystals displasses (LCD)).

MBBA (Methoxy Benzaglidine Butyle aniline).



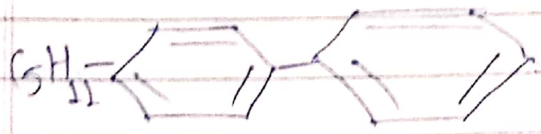
- LCP are the materials that shows properties of conventional liquids and those of a solid crystal.
- Phases of liquid crystals are called 'mesophase'

➤ Mesophase:- fundamental of a liquid crystal that induces structural order in the crystal liquid crystal molecules consiste of rigid part and one or more flexible parts. The flexible part induces fludity and rigid part is mesogen. maintance the original order of the LCP.

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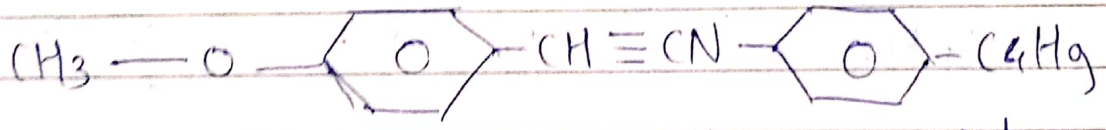
- Ex:- cellostool, esther, glycolipids, phenylbenzolyte, parapites and cellulose derivatives.

- Structure of Mesogen:-



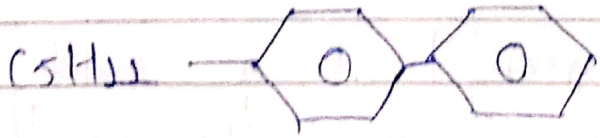
- Calamitic mesogen are rood like structure and not rood like structure are called discolamatic.

- In calamitic LCP mesogen is a rood like structure composed of two or more alafatic and aromatic range connected in one direction.
- Example:-



MIBBA (4-methoxy benzylidine-4 butyl aniline)

- Flat shaped aromatic core that makes molecde stracking one direction.



5CB (4-phenyl-4 cino biphenyl)

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- Arrangment of molecules in LCP is regular with a regular repeating pattern in all direction the molecules held in fixed position by inter-molecule forces as the T increases its molecule vibrate these vibration overcome that molecules in one place and molecule start to move.
- In liquid state these motion overcome the intermolecule forces that maintain crystaline state and molecule moves into random position without pattern in location or orientation.
- In material that forms liquid crystals inter-molecular forces in the crystaline solids not the same in all directions that is some direction forces are weeker and some are stronger
- As material is heated increased molecular motion overcomes weeker forces come but its molecule remain ~~boun~~ ~~stron~~ bound by the stronger forces thes produces the molecular arrangement that is random in some directions and regular in others.

Classification of liquid crystal polymers:-

- LCP are divided into 2 classes.

① Thermotropic.
② lyotropic

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① Thermotropic:-
- Transition liquid crystal given by changes in tempreture.

② Lyotropic:-
- phase transition we also depend concentration of the mesogen in the solvent

- thermotropic are used in tech tegnical.
- Lyotropic are used in Biology.

- Thermotropic LCP are again classified isotropic, nematic and smectic

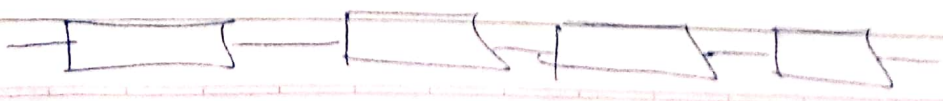
Isotropic - Randomly corelated, crystal clear, no longer range, no orientation.

Nematic - Represented by 'N', high degree molecules, polarisable, fludity of random molecules.
ex:- digital watch of base is nematic crystal.

Smectic - low tempreature, well define layers, in which molecules aligned perpendicular to the layer planes.

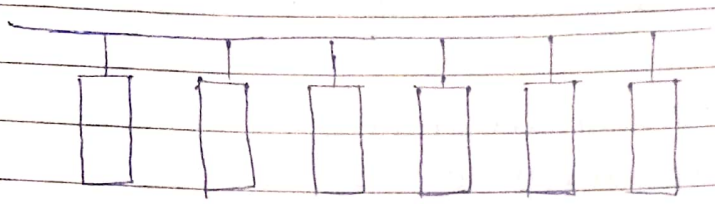
LCPs are characterized by two superimposed structural principles.

- Main chain polymers in which the rigid molecular units (mesogens) are arranged in the polymer chain and.



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- Side chain polymers which the mesogens are attached as side groups



LCP - molecular structure.



LCP :-

- Main chain polymers are not suitable for use at high temperatures.
- The side chain polymers are distinguished by particularly good electrical and optical properties which are exploited in optical ~~Strorage~~ storage or liquid crystal displays (LCD).

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LCP:-

- LCPs are based mainly on polyesters

but polyestercarbonates,
polyesteramides,
polyesterimides and
polyazomethines.

Can also be used in their production.

LCP - General properties.

- Besides the properties typical of each type of LCP, they all share the following characteristics.
  ① melt-processable.
  ② ordered structures in the melt.
  ③ highly oriented in the solid state. (self-reinforcement)
  ④ anisotropic property profile.
  ⑤ Wide variation in properties possible by selecting the appropriate LCP type.

LCP properties - 3.

- good electrical properties.
- high resistance to chemicals.
- Flammability UL 94 V-0 (0.8 mm) without flame retardant.

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- low moisture absorption.
- high resistance to stress cracking.

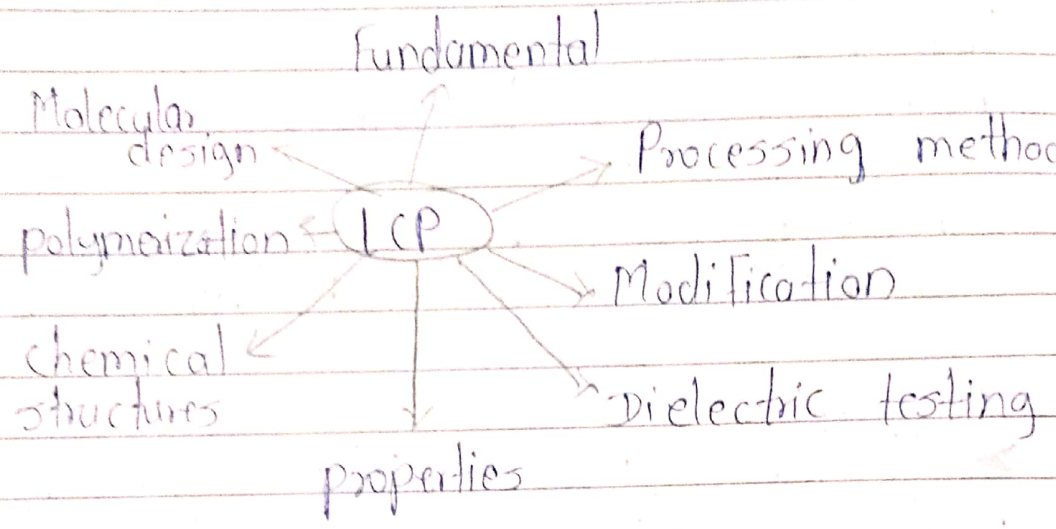
- Mainly properties, favorable and unfavorable, can be significantly influenced by the choice of processing conditions and molding design.
  - ➤ At large wall thicknesses, LCPs lose their peak properties in the longitudinal direction.
  - ➤ Shrinkage can become negative in the directions of flow.
  - ➤ Anisotropy can be reduced by adding fillers.
  - ➤ Filler content can be very high (> 40% w/w).

LCP processing.

- LCP Injection Moldings.

- The original point of using LCPs was to exploit the increase in strength which can be achieved by orientation of flow.
- The usual flat shaped design of injection moldings leads, however, to a multi-axial stress state which oriented LCP cannot cope with isotropically.

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Applications of liquid crystals:-

- Display application of liquid crystals:- The most common application of liquid crystal technology is liquid crystal displays (LCDs).
- Thermal mapping and non-destructive testing.
- Medical Uses:- cholesteric liquid crystal mixtures have also been suggested for measuring body skin tempreature, to outlines tumours etc.
- Optical Imaging and liquid Crystal Interactions with Nanostructure.
- Liquid Crystal in chromatography:-
- Liquid Crystal as solvents in spectroscopy.

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Liquid crystal polymers (LCPs):-

- The liquid crystal polymers (LCPs) are a group of chemically complete and structurally distinct materials that have unique properties and are utilized in diverse applications.

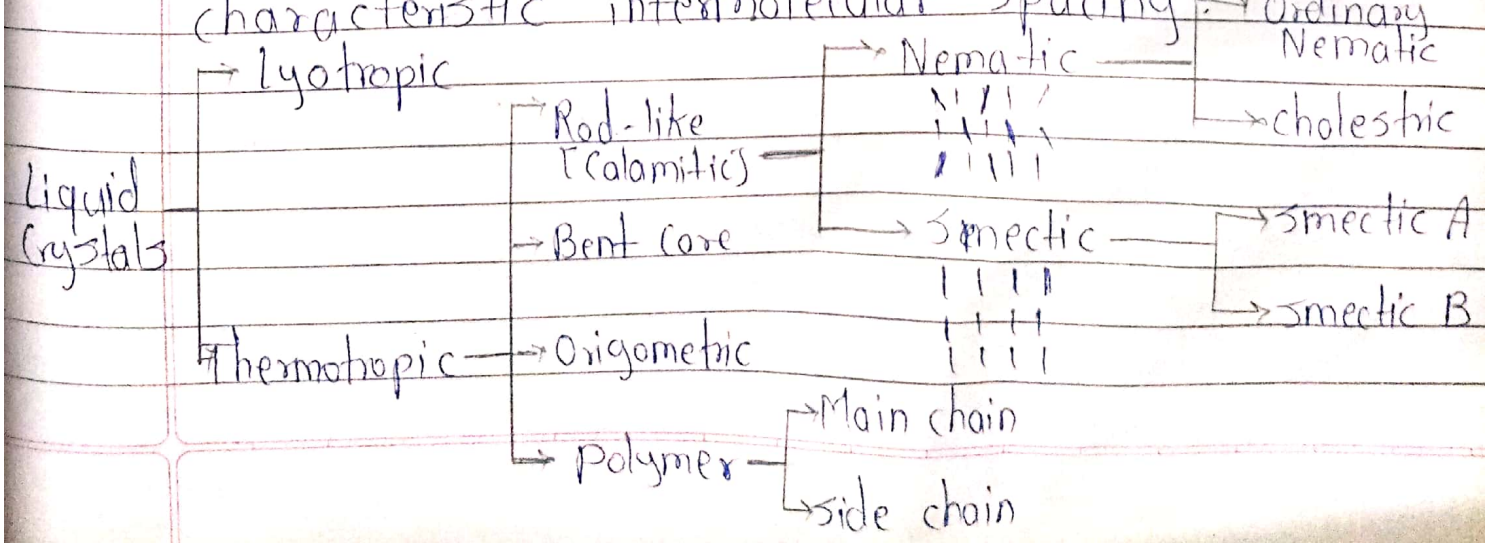
-

- LCPs are composed of extended, rod-shaped, and rigid molecules.

- In terms of molecular arrangement → these materials do not fall within any of conventional liquid amorphous, crystalline, semicrystalline classifications → as a new state of matter the liquid crystalline state, being neither crystalline nor liquid.

- In the melt (or liquid) condition → LCP molecules can become aligned in highly ordered configurations.

- As solids → this molecular alignment remains, and form in domain structures having characteristic intermolecular spacing.



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positional order + Orientational order = crystal phase.

positional order + No Orientational order = plastic phase

Varying positional order + Orientational order
= Liquid crystal phase.

No positional Order + No Orientational order
= Isotropic phase

Liquid crystals

Thermotropic

Lyotropic

High molecules (molar) mass polymers

Low molecular (molar) mass

Main chain polymers

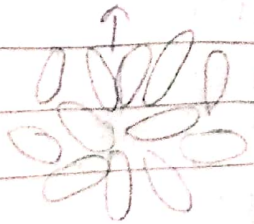
Side chain polymers

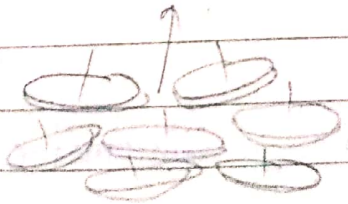
Rod-like or lath like molecules calamatic

Disc like molecules Discotic

Homo or co-polymes.

Clamitic vs. Discotic Nematics.





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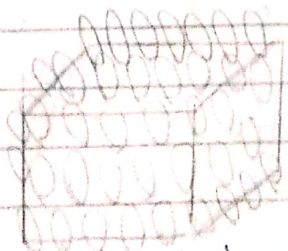
Applications:- Kevlar, the most widely used body armor is made up of intertwined liquid crystal polymers.

Liquid crystals are classified in terms of following criterion.

① Translational order / positional order.
② Bond orientational order.
③ Correlation between smectic layers.
④ With chirality.

➤ The amount of energy required to cause the phase transition is called latent heat of the transition and is useful to measure of how different the two phases are.
➤ In the case of cholesteryl myristate, the latent heat of solid to liquid crystal is 65 calories/gram, while the latent heat for liquid crystal to liquid transition is 1 calories/gram.
➤ The smallness the latent heat of liquid crystal to liquid phase transition is evidence that crystal are more similar to liquids than they are to solids.

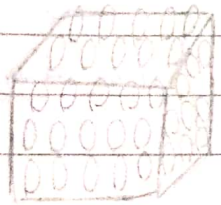
Thermotropic Liquid Crystal:-



Crystal

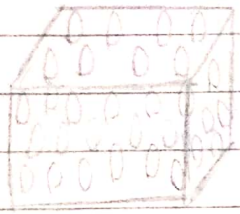
- 3-D lattice
- Orientation
- Solid
- anisotropic

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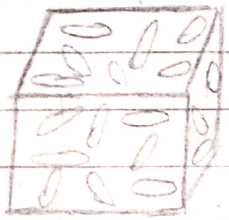


- 1-(2-)D lattice
- Orientation
- fluid
- anisotropic

liquid crystal (mesophase)



- no lattice
- Orientation.
- Fluid.
- anisotropic.



- no lattice
- no orientation.
- Fluid.
- isotropic.

Chiral liquid crystal - cholesteric.

- Also known as "chiral nematic"
- Molecules have non-symmetrical carbon atoms and thus lose mirror symmetry.
- shows a helical structure.
- In general the helical pitch of cholesteric liquid crystals is of the order of visible light's wavelength - about a few hundreds nm and so shows different colour

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Liquid crystal polymer (LCP):-

➤ Characteristics:-

- These are a class of aromatic polymer.
- Extremely unreactive and inert.
- Highly resistant to fire.

➤ Liquid crystallinity in polymers can be obtained:-

- By dissolving in a solvent. (Thermotropic).
- By heating above melting transition point. (Lyotropic).

Applications:-

- Soap
- Conducting foams.
- Heat sensitive cameras use liquid crystal screens that respond to heat.

One-dimensional translational order - smectic:-

- The word "smectic" is derived from the Greek word for soap.
- Liquid-like motion of the rods in each layer.
- No correleation of the molecular positions from one layer to the next.
- The layers can easily slide.

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- In the smectic A phase, molecules tend to be perpendicular to the smectic layers.
- In the smectic c phase, the molecules in the layers are parallel and tilted in arrangement with respect to the normal of the layers by a tilt angle θ.

Advantages of LCP:-

- High heat resistance.
- Flame ~~resis~~ retardant.
- chemical resistance.
- Dimensional stability.
- Mold ability.
- Heat aging resistance.
- Adhesion.
- Low viscosity.
- Wieldable.
- Low cost.

D = Du
R Renewable materi
o Omit derivatization st
D Degradable chemical
u - product of sat synthesis
c - catalytic agent
T - Temperature
I - In process No.
U - recycled acti. layer
E - E. fractal imaging
I - low dielectric ch
Y -

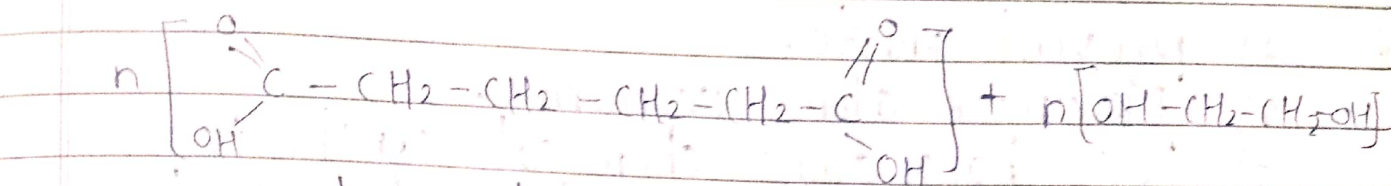
Disadvantages of LCP:-

- Form weak weld lines.
- Highly anisotropic properties.
- Drying required before processing.
- High z-axis thermal expansion coefficient.

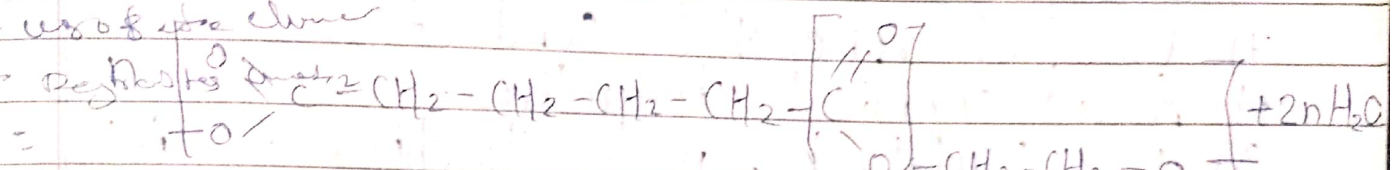
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Condensation polymerisation



hexanedioic acid — ethanediol



a polyester — ester link — water

Condensed principles of Green chemistry.

- P - prevent wastes.
- R - Renewable materials.
- O - Omit derivatization steps.
- D - Degradable chemical products.
- U - Use safe synthetic methods.
- C - Catalytic reagents.
- T - Temperature, pressure ambient
- I - In process Monitoring.
- V - Very few auxillary substances.
- E - E-factorial make in product.
- L - Low toxicity of chemical product.
- Y - yes it is safe.

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